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STA301 Lecture Summaries (Lectures 31 to 40)

Lecture 31: Sampling Distribution of $\bar{\bar{X}}$

Concept Overview: The sampling distribution of the sample mean ($\bar{\bar{X}}$) is the probability distribution of all possible sample means from all possible random samples of a given size drawn from a population.

Properties:

Mean: $\mu_{\bar{x}} = \mu$

Standard Deviation (Standard Error): $\sigma_{\bar{x}} = \sigma / \sqrt{n}$

Central Limit Theorem (CLT):

Regardless of the population's distribution, the sampling distribution of $\bar{\bar{X}}$ approaches a normal distribution as the sample size n becomes large (typically $n \geq 30$).

Implications:

CLT is the theoretical foundation for many inferential statistics techniques. It allows us to use normal distribution techniques even when population distribution is unknown.

Lecture 32: Sampling Distribution of Proportions and Difference of Means

Sampling Distribution of Proportion ($\hat{p}$):

Mean: $\mu_{\hat{p}} = p$

Standard Error: $\sigma_{\hat{p}} = \sqrt{[p(1 - p) / n]}$

Sampling Distribution of Difference Between Two Means ($\bar{\bar{X}}_1 - \bar{\bar{X}}_2$):- Mean: $\mu_1 - \mu_2$

Variance: $\sigma^2 = \sigma_1^2/n_1 + \sigma_2^2/n_2$

Standard Error: $\sqrt{[\sigma_1^2/n_1 + \sigma_2^2/n_2]}$

Conditions:

Applicable under independent random samples and sufficiently large n.

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Lecture 33: Point Estimation

Definition:

Point estimation involves providing a single value (called a point estimate) as an approximation of an unknown population parameter.

Common Point Estimators: - Sample mean ($\bar{\bar{X}}$) estimates population mean (μ)

Sample proportion ($\hat{\hat{p}}$) estimates population proportion (p)

Properties of a Good Estimator:

Unbiasedness: $E(\hat{\hat{\theta}}) = \theta$

Consistency: $\hat{\hat{\theta}} \rightarrow \theta$ as $n \rightarrow \infty$

Efficiency: Among unbiased estimators, the one with least variance is most efficient

Sufficiency: An estimator that uses all information from the sample

Lecture 34: Estimation Methods and Confidence Intervals

Methods of Estimation:

Method of Moments: Equates sample moments with theoretical population moments

Maximum Likelihood Estimation (MLE): Maximizes the probability of observing the given sample

Interval Estimation:

A confidence interval gives a range of plausible values for the parameter.

General form: Estimate ± Margin of Error

Confidence Level (e.g. 95%, 99%): Probability that the interval contains the true parameter value

Interpretation: A 95% confidence level means that 95 out of 100 such intervals will capture the true parameter.

Lecture 35: Confidence Intervals for μ and $\mu_1 - \mu_2$

Known σ: Use Z-distribution

CI for Mean: $\bar{\bar{x}} \pm Z * (\sigma / \sqrt{n})$

Unknown σ: Use t-distribution

CI for Mean: $\bar{\bar{x}} \pm t * (s / \sqrt{n})$, $df = n - 1$

CI for Difference Between Two Means:

$$(\bar{\bar{x}}_1 - \bar{\bar{x}}_2) \pm Z * \sqrt{[(\sigma_1^2 / n_1) + (\sigma_2^2 / n_2)]}$$

Interpretation:

A wider interval reflects greater uncertainty. A narrower interval suggests more precision.

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Lecture 36: Confidence Intervals for Proportions and Sample Size

Confidence Interval for a Proportion:

$\hat{p} \pm Z * \sqrt{[\hat{p}(1 - \hat{p}) / n]}$

Determining Sample Size:

To achieve a desired margin of error (E) at a specific **confidence level:**

n = $(Z^2 * \hat{p}(1 - \hat{p})) / E^2$

If $\hat{p}$ is unknown, use 0.5 to ensure maximum variability and conservative estimate.

Lecture 37: Hypothesis Testing – Introduction

Concept:

Hypothesis testing involves making decisions about population parameters using sample statistics.

Terminology:

Null Hypothesis (H_0): No effect or status quo

Alternative Hypothesis (H_1): Represents what we seek evidence for

Errors:

Type I Error (α): Rejecting H_0 when it is true

Type II Error (β): Failing to reject H_0 when it is false

Test Statistic for Mean (σ known):

Z = $(\bar{x} - \mu_0) / (\sigma / \sqrt{n})$

Decision Rule:

Compare calculated Z to critical Z-value. Reject H_0 if it falls in the rejection region.

Lecture 38: Z-tests for Means and Proportions

Two-Sample Z-Test for Means (σ known):

$Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{[(\sigma_1^2 / n_1) + (\sigma_2^2 / n_2)]}$

Z-Test for Proportions:

$Z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)]}$

Where $\hat{p} = (x_1 + x_2) / (n_1 + n_2)$

Used to compare the difference between population means or proportions.

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Lecture 39–40: t-tests and Confidence Intervals

t-Test for One Mean:

Used when σ is unknown and n is small

$t = (\bar{x} - \mu) / (s / \sqrt{n})$, $df = n - 1$

Paired t-Test:

Used for dependent samples

$t = (\bar{d} - \mu_d) / (s_d / \sqrt{n})$, where $\bar{d}$ is the mean of differences

Confidence Intervals using t:

$CI = \bar{x} \pm t\,(s / \sqrt{n})$

These tests assume normally distributed data or use the CLT for large n.

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